Experiment 3

Aim: Exploratory Data Analysis & Statistical Analysis

Objective:

1. To visualize the distribution of classes and features in the dataset.
2. To understand the spread and central tendency of data using plots.
3. To identify correlations between features using heatmaps.
4. To perform statistical hypothesis testing (e.g., t-tests) to determine if observed differences are significant (as per the requirement).

Dataset Summary Statistics (TWCS Cleaned Dataset):

Feature	Mean	Std Dev	Min	25% (Q1)	Median	75% (Q3)	Max
Word Count	19.23	9.68	5.00	12.00	18.00	24.00	68.00
Character Count	102.23	51.34	11.00	64.00	100.00	127.00	280.00
VADER Compound	+0.028	0.456	-0.991	-0.309	0.000	+0.402	+0.984
Positive Valence	0.103	0.132	0.000	0.000	0.060	0.172	0.919
Negative Valence	0.085	0.113	0.000	0.000	0.000	0.147	0.943
Neutral Valence	0.812	0.154	0.057	0.713	0.824	0.941	1.000
Exclamation Count	0.461	1.042	0.000	0.000	0.000	1.000	24.00

Detailed Steps

- Plot class balance: Use a count plot or pie chart to visualize the number of samples in each class (Neutral: 40.22%, Disappointment: 26.18%, Joy: 21.40%, Anger: 10.77%, Fear: 1.42%).

- Visualize frequency distributions of features: Use Histograms and boxplots with parametric mean and non-parametric median indicators.

- Assess Feature Correlations: Use Heatmap for Pearson linear and Spearman rank correlation matrix.

- Feature Distribution Analysis & Outlier Detection: Fit theoretical distributions (Gaussian, Log-Normal, Exponential, Poisson) to continuous/discrete variables and evaluate via Kolmogorov-Smirnov tests and Tukey IQR outlier fences.

- Hypothesis Testing: Run t-test/chi-square/ANOVA/Mann-Whitney U for mean and rank differences.

- Document EDA insights: Document distributional shapes, dispersion, skewness, and domain conclusions.

Open-Source Tools

Matplotlib, Seaborn, Plotly, SciPy, Statsmodels, Pandas, NumPy

Deliverables

A well-commented EDA Notebook including:

1. Data loading and feature preprocessing (twcs_cleaned.csv)
2. Visualizations (class balance, frequency histograms, boxplots, correlation heatmaps)
3. Statistical test implementations (Welch's t-test, ANOVA, Chi-Square, Mann-Whitney U)
4. Summary of insights and empirical domain inferences

Hypothesis test results, clearly documenting:

1. Two-Sample Welch's t-Test (Negative Complaints vs. Joy Word Count)

Hypotheses: H_0: $\mu_{negative} = \mu_{joy}$ | H_1: $\mu_{negative} \neq \mu_{joy}$
Test Statistic & Parameters: Negative (N=18,476, Mean=21.29 words, SD=9.91) vs Joy (N=10,701, Mean=16.78 words, SD=9.13), df = 23,892.4, t-statistic = **39.3351**, Cohen's d = **0.468** (Moderate effect size).

P-value and interpretation: p = **0.0000** ($p < 10^{-10}$, $\alpha = 0.05$). Since $p < 0.05$, the observed mean difference of +4.51 words is statistically significant.

Conclusion based on test result: Reject H_0. Dissatisfied customer complaints are significantly more verbose (+26.9% words) than joyful praise.

2. One-Way ANOVA & Tukey HSD (Emotion Polarity Stratification)

Hypotheses: H_0: $\mu_{anger} = \mu_{sadness} = \mu_{fear} = \mu_{neutral} = \mu_{joy}$ | H_1: At least one category differs.
Group Means (VADER Compound): Anger: -0.5543, Sadness: -0.3550, Fear: -0.1486, Neutral: +0.1356, Joy: +0.6000.

Test Statistic & P-value: F-statistic = **32,429.39**, p = **0.0000** ($p < 10^{-15}$, $\alpha = 0.05$).

P-value and interpretation: Post-hoc Tukey HSD confirms all pairwise between-group differences are statistically significant ($p < 0.001$).

Conclusion based on test result: Reject H_0. The five emotion categories form distinct, non-overlapping sentiment polarity distributions.

3. Chi-Square (χ^2) Test of Independence (Emotion vs. Time of Day)

Hypotheses: H_0: Emotion category is independent of Time of Day | H_1: Emotion distribution depends on Time of Day.

Test Statistic & Parameters: χ^2 = **56.5628**, df = 12, Cramér's V = **0.0194**.

P-value and interpretation: p = $\mathbf{9.4784 \times 10^{-8}}$ ($p < 10^{-5}$, $\alpha = 0.05$).

Conclusion based on test result: Reject H_0. Significant temporal relationship detected; angry complaints surge proportionally during Evening and Night hours.

4. Theoretical Distribution Fitting & Outlier Detection Results

Goodness-of-Fit (Kolmogorov-Smirnov Test): Log-Normal fit (Shape=0.39, Scale=22.63, KS=**0.0665**) provides the best continuous representation compared to Gaussian (KS=0.0809) and Exponential (KS=0.1640). Poisson discrete mean $\lambda = 19.23$.

Outlier Detection: Tukey's IQR fences [0.0, 42.0 words] isolate **1,799 outliers (3.60%)**. Density profiling reveals that long-tweet outliers are overwhelmingly negative complaints.

Conclusion

Experiment 3 successfully completed the Exploratory Data Analysis and Statistical Hypothesis Testing objectives. The empirical data exhibits a right-skewed Log-Normal distribution for tweet lengths (KS = 0.0665). Formal hypothesis tests confirmed with high statistical significance ($p < 10^{-10}$) that dissatisfied customer complaints demand substantially higher word counts (+4.51 words, t = 39.34) and exhibit elevated late-night volumes ($\chi^2 = 56.56$, $p < 10^{-7}$). All experimental deliverables, plots, and code implementations have been validated and documented.